

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Fourth Semester of B. Tech (EE) Examination****April-May 2018****EE205 Electrical Machines-I****Date: 07.05.2018, Monday****Time: 10.00 a.m. To 01.00 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q - 1** Draw a net sketch of a D.C. generator. Derive the e.m.f. equation of D.C. generator. **[07]**
- Q – 2.a** Define the energy and co energy in magnetic circuit. **[04]**
- Q – 2.b** **Answer any two questions.** **[10]**
- (i) Explain the basic principle of dc motor. Derive its torque equation.
 - (ii) Why a D.C. series motor should not be started at no-load?
 - (iii) 6 pole lap wound dc generator has 720 conductors; a flux of 80mweber/pole is driven at 1000rpm. Find the generated emf.
- Q - 3** **Answer any two.** **[14]**
- a. Draw the power flow diagrams of D.C. generator and D.C. motor.
 - b. Explain the necessity of starter in a D.C. motor and describe three-point starter with neat sketch.
 - c. 6 pole lap wound shunt motor has 500conductors in the armature. The resistance of armature path is 0.05ohm. The resistance of shunt field is 25ohm. Find the speed of the motor when it takes 120A from dc mains of 100V supply. Flux per pole is 2×10^{-2} Wb.

SECTION – II

- Q – 4.a** Describe the operation of a single-phase transformer; explain clearly the functions of the different parts. **[07]**
- Q – 5.a** Draw the approximate model of transformer. Also draw phasor diagram on load. **[04]**
- Q – 5.b** **Answer any two questions.** **[10]**
- a. Derive the emf equation of transformer.
 - b. Distinguish between distribution and power transformers.
 - c. The maximum flux density in the core of a 250/3000 Voltas, 50Hz single phase transformer is 1.2 Wb/m^2 . If the e.m.f per terns is 8 volts, determine (1) primary and second turns (2) area of the core.

Q – 6 Answer any two.**[14]**

- a.** State and prove the condition for maximum efficiency for single-phase transformer.
- b.** Give the Different applications of a single phase transformer.
- c.** The core of a 100KVA, 11000/550 V, 50Hz, 1-phase, core type transformer has a cross-section of 20cm×20cm. find (1) the number of H.V and L.V turns per phase and (2) the e.m.f. per turn if the maximum core density is not to exceed 1.3 tesla. Assume a stacking factor of 0.9. What will happen if its primary voltage is increased by 10% on no-load?
- d.** Explain the open circuit test on single phase transformer with neat sketch.
